Microsoft Intune First-Time Setup for Windows — User & Autopilot Enrollment (Entra-Joined)

Implementation Solution Guide

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Executive Summary — Microsoft Intune First-Time Setup for Windows

This solution takes an organization from **Intune turned on but no Windows devices under management** to **its first Windows devices enrolled two ways** — through **user-driven enrollment** (a user joins their PC to Microsoft Entra ID and it auto-enrolls into Intune) and through **basic Windows Autopilot** (a new or reset device provisions itself, zero-touch, into a managed state during the out-of-box experience). It is deliberately the **beginner on-ramp**: Windows only, **Microsoft Entra ID–joined only**, and scoped to a first pilot rather than a full production hardening.

The outcome

When you finish, you have:

- **Automatic Windows enrollment** enabled so Entra-joined PCs enroll into Intune with no manual step.
- **One Windows PC enrolled by a user** (the user-driven path) — managed and compliant.
- **One Windows PC provisioned by Autopilot** (Entra join, user-driven) — managed and compliant.
- A named, auditable set of objects (enrollment profile, ESP, device group, compliance policy) you can find, review, and clean up later.
- Read-only PowerShell scripts that verify prerequisites, capture an Autopilot hardware hash, confirm a device enrolled and is compliant, and report enrolled Windows devices by enrollment method.

Where this sits in the journey

This is the **middle step** of a three-step path, each in the same repeatable house style:

1. **Enable Microsoft Intune (Quick-Start)** — turn Intune on for the tenant (free first-mile).
2. **This solution** — get your **first Windows devices enrolled, two ways** (user-driven + basic Entra-joined Autopilot). *You are here.*
1. **Windows Autopilot** and **Cloud-First Autopilot Zero-Touch** — the deeper, production-scale Autopilot products (advanced/multi-profile Autopilot, deep Enrollment Status Page tuning, dynamic-group strategy, app packaging, Conditional Access).

Two ways to buy

- **Essentials (Self-Deploy)** — the complete click-by-click documentation, do it yourself.
- **Plus (We-Deploy)** — the documentation **plus** ready-to-run PowerShell scripts and configuration templates, **and** a done-for-you deployment: after checkout we reach out within one business day to schedule and run your basic Intune stand-up and first Windows enrollments with you.

What is intentionally out of scope

Hybrid Microsoft Entra join, advanced or multi-profile Autopilot, deep Enrollment Status Page tuning, dynamic-group strategy, co-management, and application packaging are **deliberately not** covered here. Each is its own catalog solution; this guide points to the deeper Autopilot products where those belong.

The deployment model (the same across every solution)

> ***Assess → Configure → Pilot → Validate → Expand***

1. Solution Overview — Microsoft Intune First-Time Setup for Windows

What this is

This is the **first-time, beginner setup** for managing **Windows** with Microsoft Intune. It assumes Intune is switched on for your tenant (if it is not, do the free **Enable Microsoft Intune Quick-Start** first) and gets your **first Windows PCs enrolled two complementary ways**:

- **User-driven enrollment** — a user takes an existing Windows PC, joins it to **Microsoft Entra ID**, and the PC **automatically enrolls** into Intune. Best for PCs already in someone's hands.
- **Basic Windows Autopilot (Entra-ID joined)** — a **new or freshly reset** PC provisions itself into a managed state during the out-of-box experience (OOBE), zero-touch. Best for new devices you want to ship straight to a user.

Both paths land the device in the same place: **Microsoft Entra joined, enrolled in Intune, managed, and reporting compliant.**

Scope — deliberately basic

- **Windows only.** No iOS/iPadOS, macOS, or Android here.
- **Microsoft Entra ID–joined only** (cloud join). **Hybrid Microsoft Entra join is out of scope.**
- **A first pilot**, not a production hardening. You enroll one PC each way and validate, using a small named pilot footprint you can expand later.

How Windows enrollment works (the short version)

1. **Identity first.** Every user and device is an object in **Microsoft Entra ID**. Intune trusts Entra for who the user is and which device is theirs.
1. **Licensing enables enrollment.** The enrolling user must hold an **Intune license**.
2. **Automatic enrollment** is a tenant switch: when a Windows device Entra-joins, it enrolls into Intune automatically (no separate step for the user).
1. **User-driven** = a user Entra-joins an existing PC (**Settings** → **Accounts** → **Access work or school**) and it auto-enrolls.

1. **Autopilot** = the device's **hardware hash** is registered ahead of time; at OOBE the device fetches its **deployment profile**, Entra-joins, MDM-enrolls, and an **Enrollment Status Page (ESP)** tracks setup until the device is ready.

1. **Policy and compliance** apply at every check-in; **reporting** proves state in the admin center.

Autopilot deployment modes (this solution standardizes on the first)

• **User-driven, Microsoft Entra joined** — the user drives OOBE; simplest and most common. **(Used here.)**

• Self-deploying (kiosk/shared, no user) — out of scope.

• Pre-provisioning ("white glove") for IT/partner staging — out of scope.

• Hybrid Entra join — **out of scope** (see the deeper Autopilot catalog solutions).

The objects you will create (named once, reused everywhere)

Object	Name to use	Notes
Pilot **user** group (Entra security, Assigned)	`INTBASE - Windows Users - Pilot`	Scopes auto-enrollment + compliance to pilot users.
Autopilot **device** group (Entra security, **Assigned**)	`INTBASE - Autopilot Devices`	Enrollment-time targeting; see the ownership note below.
Autopilot **deployment profile**	`INTBASE - Autopilot - Pilot`	User-driven, Entra join.
Enrollment Status Page	`INTBASE - ESP - Pilot`	Minimal; blocks nothing heavy on the pilot.
Windows enrollment restriction	`INTBASE - Windows Enrollment - Pilot`	Allow Windows; a sensible personal-device stance.
Compliance policy	`INTBASE - Compliance - Windows Pilot`	Minimal baseline so devices report compliant.

> ***Enrollment-time grouping (important):*** *the Autopilot device group* `INTBASE - Autopilot Devices` *is an **Assigned** group (not Dynamic), and the **Intune Provisioning Client** first-party service principal (AppId* `f1346770-5b25-470b-88bd-d5744ab7952c`*, shown in many tenants as *Intune Autopilot ConfidentialClient*) must be an **Owner** of it. Enrollment-time device grouping works by this principal *owning* the assigned group; without the owner the group stays empty at enrollment and nothing deploys. This is covered as a step and a prerequisite.*

Benefits

• **Two enrollment paths from one setup** — cover PCs already in use *and* brand-new devices.

• **Zero-touch for new devices** — ship straight to users; no technician imaging.

• **Consistent, named objects** — easy to audit and clean up.

• **Pilot-first** — prove it works on one device each way before you expand.

Limitations (be honest with buyers)

- Windows editions must be **Pro, Enterprise, or Education** — **not Home**.
- Autopilot devices must be **registered** (hardware hash) before the profile applies.
- OOBE requires **internet** and reachable Entra/Intune endpoints.
- **Entra-joined only** — Hybrid Entra join, advanced/multi-profile Autopilot, and deep ESP tuning are

out of scope (deferred to the deeper Autopilot catalog solutions).

Government (GCC / GCC High / DoD) note

US Government tenants use **different portals and endpoints**: Intune admin center at `https://intune.microsoft.us`, Microsoft Entra at `https://entra.microsoft.us`, sign-in at `login.microsoftonline.us`, and Microsoft Graph at `https://graph.microsoft.us` (`Connect-MgGraph -Environment USGov`, or `USGovDoD` for DoD). The steps are otherwise the same, but **Autopilot and enrollment feature timing can differ in GCC High/DoD** — where a difference is uncertain, follow the Commercial path and **verify for GCC High** against Microsoft's US sovereign-cloud documentation rather than assuming parity.

Auth and connections used

- **Microsoft Intune admin center** (`https://intune.microsoft.com`) — enrollment settings, Autopilot,

ESP, compliance.

- **Microsoft Entra admin center** (`https://entra.microsoft.com`) — licensing, groups, auto-enrollment

scope, service-principal ownership.

- **Microsoft Graph PowerShell** (optional) — the included scripts use it **read-only** to verify and

report state (remediation defaults to a safe `-WhatIf` preview).

- You act as a **Global Administrator** (first-time tenant changes) or an **Intune Administrator**.

The standard deployment model

> ***Enable auto-enrollment → create pilot groups → configure user-driven enrollment → register + build basic Autopilot (profile + ESP) → enroll one PC each way → validate → expand.*** *Every step follows the repeatable pattern:* ***Assess → Configure → Pilot → Validate → Expand.***

2. Requirements — Microsoft Intune First-Time Setup for Windows

Confirm each requirement below **before** you start. For every item: what it is, why it is required, how to verify it, and how to put it in place if it is missing. Do not assume anything already exists.

1. Intune is enabled for the tenant

• **Description:** Intune is switched on and is the MDM authority for the tenant.

• **Why required:** There is nothing to enroll a Windows device into until Intune is active. This solution is the step *after* enablement.

• **How to verify:** Sign in at `https://intune.microsoft.com`; **Tenant administration** → **Tenant status** shows the tenant active. **Devices** → **Enrollment** is reachable.

• **How to implement if missing:** Complete the **Enable Microsoft Intune (Quick-Start)** first — it turns Intune on for a new or existing tenant. This guide does not reproduce that first-mile detail.

2. Global Administrator (or Intune Administrator) access

• **Description:** A role with permission to configure enrollment, Autopilot, and groups.

• **Why required:** Setting the auto-enrollment scope, creating Autopilot profiles/ESP, and making a service principal a group owner are tenant-level actions.

• **How to verify:** In **Entra admin center** → **Roles & admins**, confirm your account holds **Global Administrator **or** Intune Administrator**. (Making the Intune Provisioning Client an *owner* of a group needs sufficient Entra rights — Global Administrator, or a role that can edit group owners.)

• **How to implement if missing:** Have an existing Global Administrator assign the role.

3. Intune license assigned to the enrolling user

• **Description:** An Intune entitlement, standalone or bundled in **Enterprise Mobility + Security (EMS)** or a **Microsoft 365** plan (Business Premium, E3/E5).

• **Why required:** A user cannot enroll a device unless they are licensed for Intune. This is the single most common reason enrollment fails.

• **How to verify:** In **Entra admin center** → **Billing** → **Licenses**, confirm you own Intune-bearing licenses with free seats, then confirm the pilot user is assigned one under the user's **Licenses** blade.

• **How to implement if missing:** Purchase/trial an Intune-bearing plan and assign a license to each pilot user (directly or via the `INTBASE - Windows Users - Pilot` group). A **30-day trial** is sufficient — *verify current trial availability in your tenant*.

4. A pilot user (cloud identity)

• **Description:** A cloud user who will perform user-driven enrollment and sign in on the Autopilot device.

• **Why required:** Enrollment binds a device to a signed-in, licensed user.

• **How to verify:** In **Entra admin center** → **Users**, confirm a non-admin cloud user exists, is licensed (item 3), and can sign in.

• **How to implement if missing:** Create one under **Users** → **New user**, name it clearly (e.g. `Windows Pilot User 01`), assign an Intune license, and set a password for the test devices.

5. Pilot Entra security groups

• **Description:** Two Microsoft Entra security groups: a **user** group and an Autopilot **device** group.

• **Why required:** Auto-enrollment scope and compliance target the **user** group (`INTBASE - Windows Users - Pilot`); the Autopilot profile and ESP target the **device** group (`INTBASE - Autopilot Devices`).

• **How to verify:** In **Entra admin center** → **Groups**, confirm both groups exist with the exact names.

• **How to implement if missing:** Create them in the implementation steps. The user group is **Assigned** with the pilot user as a member; the device group is **Assigned** (see item 6).

6. Intune Provisioning Client is an OWNER of the Autopilot device group

• **Description:** The first-party service principal **Intune Provisioning Client** (AppId `f1346770-5b25-470b-88bd-d5744ab7952c`, often displayed as *Intune Autopilot ConfidentialClient*) must be an **Owner** of the **Assigned** device group `INTBASE - Autopilot Devices`.

• **Why required:** Enrollment-time device grouping works by this principal *owning* the assigned group. Without the owner, the group stays empty at enrollment and the Autopilot profile/ESP never deploy.

• **How to verify: Entra** → **Groups** → `INTBASE - Autopilot Devices` → **Owners** lists a principal whose AppId is `f1346770-5b25-470b-88bd-d5744ab7952c`. Search by **AppId**, not the display name (which varies by tenant).

• **How to implement if missing:** Add it as an owner (implementation Step 6). If the service principal does not exist yet, provision it by **opening the Intune admin center once as a Global Administrator**, then retry. *(Tenant-verifiable: the display name varies; match on the AppId.)*

7. Supported Windows test devices

• **Description:** For user-driven — a Windows 10 (1809+) or Windows 11 PC that can be Entra-joined. For Autopilot — a **new or factory-reset** PC of the same editions.

• **Why required:** To prove both enrollment paths. Autopilot needs a device at the OOBE screen.

• **How to verify:** Confirm the edition is **Pro, Enterprise, or Education** (`winver`) — **not Home** — and the device can reach the internet.

• **How to implement if missing:** Use spare PCs or cloud VMs. For Autopilot, plan to reset the device so

it starts at OOBE.

8. Autopilot hardware hash for the Autopilot device

• **Description:** The device's Autopilot hardware identity (4K hardware hash) registered in Intune.

• **Why required:** Autopilot only applies a profile to a **registered** device.

• **How to verify: Intune** → **Devices** → **Enrollment** → **Devices** (Windows Autopilot devices) lists the

device serial after import.

• **How to implement if missing:** Collect it with the included `INTBASE-Get-AutopilotHash.ps1` (or

Microsoft's `Get-WindowsAutopilotInfo`) and import the CSV, **or** have your OEM/reseller register the device to your tenant at purchase.

9. Network access from devices

• **Description:** Devices must reach Microsoft Intune, Entra, and Windows notification endpoints.

• **Why required:** Enrollment, OOBE provisioning, and check-in are cloud calls.

• **How to verify:** From a device, reach `https://login.microsoftonline.com` and

`https://enrollment.manage.microsoft.com` (Commercial). On restricted networks, allow the documented Microsoft Intune/Autopilot endpoints — *verify against current Microsoft endpoint documentation*.

• **How to implement if missing:** Allow the documented Microsoft 365/Intune URL & IP ranges through the

proxy/firewall.

Government (GCC / GCC High / DoD) variance

For US Government tenants the portals and endpoints differ — Intune `https://intune.microsoft.us`, Entra `https://entra.microsoft.us`, sign-in `login.microsoftonline.us`, Graph `https://graph.microsoft.us` (`Connect-MgGraph -Environment USGov`). The Intune Provisioning Client AppId is the **same**. Autopilot and enrollment **feature timing can differ in GCC High/DoD**; where uncertain, follow the Commercial path and **verify for GCC High** against Microsoft's US sovereign-cloud documentation.

Global Requirements — Intune Solutions (category-level)

These requirements apply to **every** solution in the **Intune Solutions** category. They are published once at the category level so individual listings stay small. Confirm these before deploying *any* Intune solution. Each entry: **Description · Why · Verify · Implement-if-missing.**

G1. Microsoft Entra ID tenant

Description: An active Microsoft Entra ID (Azure AD) tenant. **Why:** It is the identity backbone for every cloud and Intune-managed resource. **Verify:** Sign in to `https://entra.microsoft.com` as an administrator. **Implement:** Create a Microsoft 365/Azure subscription, which provisions a tenant.

G2. Microsoft Intune subscription

Description: An Intune-bearing license plan. **Why:** Intune delivers all configuration, compliance, and security policies. **Verify:** `https://intune.microsoft.com` loads; **Tenant administration** → **Tenant status** shows active. **Implement:** Assign Microsoft 365 E3/E5, Business Premium, or standalone Intune Plan 1.

G3. Administrative roles

Description: Appropriate Entra/Intune admin roles for the operator. **Why:** Creating policies and enrollment settings requires elevated rights. **Verify:** Entra → **Roles and administrators**; confirm **Intune Administrator**. **Implement:** Assign **Intune Administrator**; reserve Global Administrator for tenant-wide changes.

G4. Device enrollment in Intune

Description: Targeted Windows devices are enrolled and manageable. **Why:** Policies only reach enrolled devices. **Verify:** Intune → **Devices** → **All devices**; the device appears and is **compliant**. **Implement:** Enable **Automatic Enrollment** and join devices to Entra ID.

G5. Pilot security group

Description: A Microsoft Entra security group for pilot/targeted users or devices. **Why:** Every solution is piloted before fleet rollout. **Verify:** Entra → **Groups**; the pilot group exists. **Implement:** Entra → **Groups** → **New group** → **Security** → add members.

G6. Supported Windows baseline

Description: Windows 10 21H2+ or Windows 11 with current updates. **Why:** Modern Intune features assume a patched, supported OS. **Verify:** `winver`, or Intune → device → **Hardware**. **Implement:** Deploy **Update rings** or upgrade to Windows 11.

G7. Network access to Microsoft cloud endpoints

Description: Outbound HTTPS to Microsoft 365, Entra, and Intune service URLs. **Why:** Enrollment, policy sync, and authentication depend on these endpoints. **Verify:** From a device, reach `https://login.microsoftonline.com` and `https://enrollment.manage.microsoft.com`. **Implement:** Allow Microsoft's published Office 365/Intune URL & IP ranges through proxy/firewall.

The standard deployment model (all solutions): Assess → Configure → Pilot → Validate → Expand.

3. Implementation Guide — Microsoft Intune First-Time Setup for Windows

Follow these steps **in order**. Navigation is in the **Microsoft Intune admin center** (`https://intune.microsoft.com`) and **Microsoft Entra admin center** (`https://entra.microsoft.com`) as noted. *(Government tenants: use `intune.microsoft.us` and `entra.microsoft.us`.)*

> ***Model:*** *enable auto-enrollment → build pilot groups → user-driven enrollment → basic Entra-joined Autopilot (profile + ESP) → enroll one PC each way → validate → expand. Naming convention:* `<Purpose> - <Scope/Platform> - <Stage>`.

Phase A — Configure (tenant-wide switches + pilot scope)

Step 1 — Enable automatic Windows enrollment

1. In the **Microsoft Intune admin center**, go to **Devices** → in the left menu under **Device onboarding** **select** Enrollment**.

1. Open the **Windows** tab → **Automatic Enrollment**.

2. Set **MDM user scope** to **Some**, then **Select groups** and choose `INTBASE - Windows Users - Pilot` (you create this group in Step 3; return here if needed). *(Setting scope to **All** enrolls every Entra-joined Windows device tenant-wide — for a first pilot, keep it to the pilot group.)*

1. Leave **MAM user scope** as **None** for this solution. Click **Save**.

Step 2 — Set a first-run branding (optional but recommended)

1. Open `https://entra.microsoft.com` → **Home** → search **Company branding** (or **Entra ID → Company branding**).

1. Click **Customize** (or **Add**) on the default sign-in experience; add your logo, background, and sign-in text; **Save**. *(Branding makes the Autopilot OOBE and Windows sign-in show your organization, so pilot users trust the flow.)*

Step 3 — Create the pilot USER group

1. In `https://entra.microsoft.com` go to **Groups** → **All groups** → **New group**.

2. **Group type:** Security. **Group name:** `INTBASE - Windows Users - Pilot`.

3. **Membership type: Assigned**. Under **Members**, add your pilot user (e.g. `Windows Pilot User 01`). Click **Create**.

1. Return to **Step 1** if you deferred it and select this group for the **MDM user scope**.

Step 4 — Create the pilot Windows compliance policy

1. In the **Intune admin center**, go to **Devices** → **Compliance** (under **Manage devices**) →

Create policy.

1. **Platform:** Windows 10 and later. Click **Create**.
2. **Name:** `INTBASE - Compliance - Windows Pilot`. Click **Next**.
3. Under **Compliance settings**, expand **Device Health** and set **Require BitLocker** = **Not

configured **for the pilot (keep it minimal); expand** System Security **and set** Require a password to unlock mobile devices **=** Require**,** Minimum password length **=** 6**. *(Keep the baseline small — deep compliance hardening is a separate catalog solution.)* Click** Next**.

1. **Actions for noncompliance:** leave the default **Mark device noncompliant** immediately. Click

Next.

1. **Assignments** → **Add groups** → `INTBASE - Windows Users - Pilot`. Click **Next** → **Create**.

Phase B — User-driven enrollment (an existing PC)

Step 5 — End-user: Entra-join the PC (auto-enrolls into Intune)

Perform this on the **user-driven** pilot PC, signed in as a local user:

1. Open **Settings** → **Accounts** → **Access work or school**.
2. Click **Connect**. On the dialog, click **Join this device to Microsoft Entra ID** (the small link

under the email box — **not** the email box itself, which only adds a work account).

1. Sign in with the pilot user's work account (e.g. `pilotuser01@contoso.com`) and complete the prompts.

Accept the confirmation to join.

1. When it finishes, the PC is **Entra joined** and, because automatic enrollment is on (Step 1), it

auto-enrolls into Intune.

1. Verify on the device: **Settings** → **Accounts** → **Access work or school** shows the account with an

Info button; open **Info** and confirm the managed connection and click **Sync** to force a check-in. *(Optionally run `dsregcmd /status` and confirm `AzureAdJoined : YES` and an `MDMUrl`.)*

Phase C — Basic Windows Autopilot (Entra-ID joined)

Step 6 — Create the Autopilot DEVICE group and set the required owner

1. In `https://entra.microsoft.com` go to **Groups** → **All groups** → **New group**.
2. **Group type:** Security. **Group name:** `INTBASE - Autopilot Devices`. **Membership type:**

Assigned (not Dynamic). Click **Create**.

1. Open the group → **Owners** → **Add owners**. In the search box type the **AppId**

`f1346770-5b25-470b-88bd-d5744ab7952c` (the **Intune Provisioning Client**, sometimes shown as *Intune Autopilot ConfidentialClient*). Select it and click **Select**.

> ***Why:*** *enrollment-time device grouping requires this first-party service principal to* ***own*** *the assigned group. Without the owner, devices never land in the group at enrollment and the profile/ESP do not deploy. If the principal does not appear, open the Intune admin center once as a* ***Global Administrator*** *to provision it, then retry. Search by* ***AppId****, not the display name.*

Step 7 — Register the Autopilot device (hardware hash)

1. On the pilot Autopilot PC (elevated PowerShell), run `INTBASE-Get-AutopilotHash.ps1` to export

the hardware hash CSV. *(Or have your OEM/reseller register the device to your tenant at purchase — no import needed.)*

1. In the **Intune admin center**, go to **Devices** → **Enrollment** → open the **Windows** tab →

Devices (under **Windows Autopilot**) → **Import**.

1. Upload the CSV and wait for the device serial to appear. Click **Sync** and refresh if it is not

listed yet (registration can take a few minutes).

1. Add the device to the group: **Entra → Groups → `INTBASE - Autopilot Devices` → Members → Add

members**, and add the imported Autopilot device (by serial). *(At scale a dynamic ZTDId group is used instead — deferred to the deeper Autopilot solution; for a first pilot, add the one device explicitly.)*

Step 8 — Create the Autopilot deployment profile

1. Go to **Devices** → **Enrollment** → **Windows** tab → **Deployment Profiles** (under **Windows

Autopilot**)** → Create profile → Windows PC**.

1. **Name:** `INTBASE - Autopilot - Pilot`. **Description:** first-time user-driven Entra-joined pilot.

Set **Convert all targeted devices to Autopilot** = **No** (explicit registration only). Click **Next**.

1. **Out-of-box experience (OOBE):**

• **Deployment mode: User-driven**.

• **Join to Microsoft Entra ID as: Microsoft Entra joined**.

• **Microsoft Software License Terms:** Hide. **Privacy settings:** Hide. **Hide change account

options:** Hide.

• **User account type: Standard**.

• **Allow pre-provisioned deployment: No** (out of scope). **Language / keyboard:** as you prefer.

Click **Next**.

1. **Assignments** → **Add groups** → `INTBASE - Autopilot Devices`. Click **Next** → **Create**.

Step 9 — Create the Enrollment Status Page (ESP)

1. Go to **Devices** → **Enrollment** → **Windows** tab → **Enrollment Status Page** (under **Windows

Autopilot**)** → Create**.

1. **Name:** `INTBASE - ESP - Pilot`. Click **Next**.

2. **Settings:**

• **Show app and profile configuration progress: Yes**.

• **Block device use until all apps and profiles are installed: Yes**.

• **Allow users to reset device if installation error occurs: Yes**.

• **Allow users to use device if installation error occurs: Yes** (keeps the pilot from getting stuck).

• **Block device use until required apps are installed:** leave as **All** but do **not** require heavy apps on the pilot (keep the ESP light — deep ESP tuning is a separate solution). Click **Next**.

1. **Assignments** → **Add groups** → `INTBASE - Autopilot Devices`. Click **Next** → **Create**.

Step 10 — Set a Windows enrollment restriction (sensible default)

1. Go to **Devices** → **Enrollment** → **Windows** tab → **Enrollment device platform restrictions** → open the **Windows restrictions** default (or **Create restriction**).

1. If creating: **Name:** `INTBASE - Windows Enrollment - Pilot`; **Platform:** Windows; **Personally owned devices: Block **(corporate-only for the pilot). **Assign to** `INTBASE - Windows Users - Pilot`**.** Save/Create**. *(This keeps the pilot to corporate devices; you can relax it later.)*

Phase D — Pilot

Step 11 — Provision the Autopilot pilot device

1. **Factory-reset** the registered Autopilot PC (or use a new one already registered) so it starts at **OOBE**.

1. Connect it to the internet at the region/keyboard screens.

2. The Autopilot branding appears; the pilot user signs in with their **work account**.

3. The device **Entra-joins**, **MDM-enrolls**, and the `INTBASE - ESP - Pilot` page runs until the device is ready.

Phase E — Validate & Expand

Step 12 — Validate both devices

1. Run `INTBASE-Validate-Enrollment.ps1` on each pilot PC (or by device name via Graph) → confirms the device is Entra-joined, Intune-enrolled, and compliant.

1. Run `INTBASE-Report-Enrollment.ps1` → lists enrolled Windows devices with their **enrollment method** (user-driven vs Autopilot).

1. Complete the **Testing Guide** (Section 5) and confirm every PASS/FAIL gate before expanding.

Step 13 — Expand (when ready)

1. Add more pilot users to `INTBASE - Windows Users - Pilot` (or widen the auto-enrollment scope) for

user-driven PCs.

1. Register more Autopilot devices and add them to `INTBASE - Autopilot Devices` (OEM registration at

purchase is ideal at scale).

1. Re-validate each wave. For production-scale Autopilot (dynamic-group strategy, multi-profile,

pre-provisioning, deep ESP tuning, app packaging, Conditional Access), move to the deeper Autopilot catalog solutions.

Order of operations

`Auto-enrollment → pilot user group → compliance → user-driven join → Autopilot device group (+ SP owner) → register device → deployment profile → ESP → enrollment restriction → provision → validate → expand`

5. Testing & Validation — Microsoft Intune First-Time Setup for Windows

Do not declare success until **both** enrollment paths pass. The exit criterion for this solution is: **a user-enrolled Windows PC AND an Autopilot-provisioned Windows PC are both Microsoft Entra joined, Intune-enrolled, managed, and reporting compliant.**

Run the included scripts to gather evidence, and record each gate PASS/FAIL.

Test 1 — User-driven enrolled device lands managed & compliant

Setup: the PC from implementation Step 5 (`Access work or school → Join this device to Microsoft Entra ID`).

#	Check	How	PASS criteria
1.1	Device is Entra joined	On device: `dsregcmd /status`	`AzureAdJoined : YES`
1.2	Device is Intune-enrolled	On device: `dsregcmd /status` shows an `MdmUrl`; or Intune → **Devices** → **All devices** lists it	Device present, **Managed by: Intune (MDM)**
1.3	Enrollment method is correct	Intune → device → **Overview** / run `INTBASE-Report-Enrollment.ps1`	Method shows user-driven (Entra join), not Autopilot

#	Check	How	PASS criteria
1.4	Baseline compliance applied	Intune → device → **Device compliance**	`INTBASE - Compliance - Windows Pilot` evaluated
1.5	Device reports compliant	Intune → device → **Overview** → **Compliance**	**Compliant** (allow one check-in cycle)

Gate 1 — PASS only if 1.1–1.5 all pass.

Test 2 — Autopilot device provisions and lands managed & compliant

Setup: the registered device from Steps 7–11, factory-reset and taken through OOBE.

#	Check	How	PASS criteria
2.1	Device is registered for Autopilot	Intune → **Devices** → **Enrollment** → **Windows** → **Devices**	Serial present, profile `INTBASE - Autopilot - Pilot` assigned
2.2	Device is in the assigned group	Entra → **Groups** → `INTBASE - Autopilot Devices` → **Members**	Device is a member
2.3	OOBE applied the profile	At OOBE	Org branding + Autopilot flow shown; user signs in once
2.4	ESP completed	On device during setup	`INTBASE - ESP - Pilot` reaches completion, device usable
2.5	Device is Entra joined + Intune-enrolled	`dsregcmd /status`; Intune → **All devices**	`AzureAdJoined : YES`, **Managed by: Intune (MDM)**
2.6	Enrollment method is correct	`INTBASE-Report-Enrollment.ps1`	Method shows **Windows Autopilot**
2.7	Device reports compliant	Intune → device → **Overview** → **Compliance**	**Compliant**

Gate 2 — PASS only if 2.1–2.7 all pass.

Test 3 — Automated validation (scripts)

1. Run `INTBASE-Preflight.ps1` — confirms licensing, roles, MDM authority, auto-enrollment scope, and

that the Intune Provisioning Client owns `INTBASE - Autopilot Devices`. Expect all **PASS**.

1. Run `INTBASE-Validate-Enrollment.ps1 -DeviceName <name>` for each pilot PC — confirms Entra join,

Intune enrollment, and compliant state. Expect **PASS** for both devices.

1. Run `INTBASE-Report-Enrollment.ps1` — the two pilot devices appear, one **User-driven** and one

Autopilot.

Gate 3 — PASS only if Preflight is all-PASS and both devices validate.

Overall exit criteria

- **Gate 1 PASS** (user-driven device managed & compliant) **and**
- **Gate 2 PASS** (Autopilot device managed & compliant) **and**
- **Gate 3 PASS** (scripts confirm both).

Only when all three gates pass is the first-time Windows setup complete. Then expand per implementation Step 13.

6. Troubleshooting — Microsoft Intune First-Time Setup for Windows

Work top-down: confirm licensing/roles first, then enrollment scope, then the enrollment-time grouping owner, then device-side state. Run `INTBASE-Preflight.ps1` to catch the common tenant-side causes fast.

User-driven enrollment issues

"Access work or school" only adds an account — the PC does not Entra-join

- **Cause:** the user clicked the email box (adds a work account) instead of **Join this device to

Microsoft Entra ID**.

- **Fix:** in the Connect dialog, click the link **Join this device to Microsoft Entra ID** *under* the

email box, then sign in with the work account.

PC Entra-joins but never appears in Intune

- **Cause 1:** the user is **not licensed** for Intune → **Fix:** assign an Intune license (via

`INTBASE - Windows Users - Pilot`) and re-sync.

- **Cause 2:** automatic enrollment scope excludes the user → **Fix:** **Devices → Enrollment → Windows →**

Automatic Enrollment**, confirm** MDM user scope **includes** `INTBASE - Windows Users - Pilot` **(or set** All**).

• **Cause 3:** Windows **Home** edition → **Fix:** enrollment requires Pro/Enterprise/Education; upgrade the edition.

• **Recheck:** on device `dsregcmd /status` → `AzureAdJoined : YES` and a populated `MdmUrl`; force **Settings** → **Access work or school** → **Info** → **Sync**.

Device shows enrolled but not compliant

• **Cause:** first compliance evaluation has not completed, or a setting in `INTBASE - Compliance - Windows Pilot` is not yet met.

• **Fix:** allow one check-in cycle; open Intune → device → **Device compliance** to see which setting failed and remediate. For the pilot, keep the baseline minimal.

Autopilot issues

Device group is empty after enrollment / profile & ESP never deploy

• **Cause (most common):** the **Intune Provisioning Client** service principal (AppId `f1346770-5b25-470b-88bd-d5744ab7952c`) is **not an Owner** of the **Assigned** group `INTBASE - Autopilot Devices`. Enrollment-time device grouping requires this principal to *own* the group; without it the device never lands in the group and nothing deploys.

• **Fix: Entra** → **Groups** → `INTBASE - Autopilot Devices` → **Owners** → **Add owners**, search by **AppId** `f1346770-5b25-470b-88bd-d5744ab7952c`, add it. If it does not appear, open the Intune admin center once as a **Global Administrator** to provision the principal, then retry. Re-provision the device.

OOBE shows a normal Windows setup, not the Autopilot/branded flow

• **Cause 1:** the device is **not registered** for Autopilot → **Fix:** **Devices → Enrollment → Windows → Devices**; import the hash CSV (from `INTBASE-Get-AutopilotHash.ps1`) and wait for the serial to appear;** Sync**.

• **Cause 2:** no profile assigned → **Fix:** confirm `INTBASE - Autopilot - Pilot` is assigned to `INTBASE - Autopilot Devices` and the device is a **member** of that group.

• **Cause 3:** the device already passed the network screen before the profile was ready → **Fix:** factory-reset and restart OOBE after registration/assignment shows on the device record.

ESP hangs / device blocked at "Setting up your device"

• **Cause:** `INTBASE - ESP - Pilot` requires apps that are slow or failing to install.

• **Fix:** keep the pilot ESP light — do not require heavy apps; ensure **Allow users to use device if installation error occurs = Yes** so the pilot is not stuck. Deep ESP tuning is a separate solution.

Imported device does not show up

- **Cause:** registration propagation delay, or a malformed CSV.
- **Fix:** click **Sync** and wait a few minutes; confirm the CSV has the required columns (see `autopilot-hardware-hash.csv` template) with no extra headers. Re-run `INTBASE-Get-AutopilotHash.ps1` elevated if the export was empty.

Script issues

`Connect-MgGraph` *fails or lacks permission*

- **Fix:** consent to the read scopes on first connect. For Government tenants use `Connect-MgGraph -Environment USGov` (or `USGovDoD`). Preflight/report scripts are read-only.

`INTBASE-Get-AutopilotHash.ps1` *produces no file*

- **Fix:** run it **elevated**; it installs and calls Microsoft's `Get-WindowsAutopilotInfo`. Confirm the device has internet access to install the script.

`INTBASE-Remediation.ps1` *reports it "would" change something but did not*

- **Expected:** it defaults to `-WhatIf` (preview). Review the preview, then re-run with `-Apply` to make the change. It only touches the auto-enrollment scope / the group owner it is scoped to.

Government (GCC / GCC High / DoD)

- Use `intune.microsoft.us` / `entra.microsoft.us`; scripts use `Connect-MgGraph -Environment USGov`.
- The Intune Provisioning Client AppId is the same. If an Autopilot/enrollment feature behaves differently than documented here, treat it as a **GCC High parity/timing** difference and **verify against Microsoft's US sovereign-cloud documentation** rather than assuming Commercial behavior.

7. Best Practices — Microsoft Intune First-Time Setup for Windows

Scope discipline (keep the first setup small)

- **Pilot first, always.** Enroll one PC each way, validate against the gates, then expand. Resist the urge to widen the auto-enrollment scope to **All** before the pilot passes.
- **Keep the ESP light on the pilot.** Do not require heavy apps in `INTBASE - ESP - Pilot` — a blocked ESP is the most common reason a first Autopilot run appears to "hang". Deep ESP tuning belongs to the advanced Autopilot solution.
- **Entra-joined only.** Do not mix in Hybrid Entra join here; it changes the identity model and is a

separate solution.

Naming and hygiene

- **Use the exact object names** throughout: `INTBASE - Windows Users - Pilot`, `INTBASE - Autopilot Devices`, `INTBASE - Autopilot - Pilot`, `INTBASE - ESP - Pilot`, `INTBASE - Windows Enrollment - Pilot`, `INTBASE - Compliance - Windows Pilot`. Consistent names make the setup auditable and easy to clean up.
- **One purpose per group.** Keep the pilot **user** group and the Autopilot **device** group separate — auto-enrollment/compliance target users; Autopilot profile/ESP target devices.

The enrollment-time grouping rule (do not skip)

- The Autopilot device group is **Assigned**, and the **Intune Provisioning Client** service principal (AppId `f1346770-5b25-470b-88bd-d5744ab7952c`) must **own** it. Verify ownership before provisioning — it is the difference between a working zero-touch device and an empty group where nothing deploys.

Enrollment method choice

- **User-driven** for PCs already in someone's hands (fastest to prove the tenant works).
- **Autopilot** for new/reset devices you want to ship zero-touch. Prefer **OEM/reseller registration at purchase** over manual hash import once you move past the pilot.

Licensing and identity

- Assign the Intune license **via the pilot group** so new pilot users inherit it automatically.
- Use a **dedicated pilot user**, not an admin account, so the experience matches a real end user.

Validation and evidence

- Keep the scripts **read-only** in normal use; run `INTBASE-Remediation.ps1` only with `-WhatIf` first and read the preview before applying.
- Save the output of `INTBASE-Report-Enrollment.ps1` as your record that both paths landed managed and compliant.

Government tenants

- Use the `*.us` portals and `Connect-MgGraph -Environment USGov` for scripts. Treat Autopilot/enrollment feature timing in GCC High/DoD as **verify-before-relying**, not assumed parity.

When to move up

• Ready for dynamic-group strategy, multi-profile Autopilot, pre-provisioning ("white glove"), deep ESP tuning, app packaging, or Conditional Access? Those are **out of scope here** by design — step up to the **Windows Autopilot** and **Cloud-First Autopilot Zero-Touch** solutions, which build directly on this foundation.

8. Deliverables Summary — Microsoft Intune First-Time Setup for Windows

What you receive

Documentation (both variants)

• **Implementation Solution Guide (PDF + Word)** — the complete, click-by-click solution:

• Executive Summary

• Solution Overview (what it is, how Windows enrollment works, scope, objects created)

• Requirements (licensing, roles, Entra, groups, service-principal ownership, devices, network — each with Description · Why · Verify · Implement-if-missing)

• Implementation Guide (ordered: enable auto-enrollment → pilot groups → user-driven enrollment → basic Entra-joined Autopilot → pilot → validate → expand)

• Testing & Validation (PASS/FAIL gates for both enrollment paths)

• Troubleshooting

• Best Practices

• Intune Solutions category global requirements

PowerShell scripts (Plus / We-Deploy variant only)

Five Microsoft Graph / Windows scripts, read-only by default and cloud-aware (`Connect-MgGraph -Environment Global | USGov | USGovDoD`):

• `INTBASE-Preflight.ps1` — verify licensing, roles, MDM authority, Windows auto-enrollment scope, and that the Intune Provisioning Client owns `INTBASE - Autopilot Devices`.

• `INTBASE-Get-AutopilotHash.ps1` — collect/import a device's Autopilot hardware hash (CSV).

• `INTBASE-Validate-Enrollment.ps1` — confirm a device is Entra-joined, Intune-enrolled, and compliant.

• `INTBASE-Report-Enrollment.ps1` — list enrolled Windows devices with enrollment method (user-driven vs Autopilot).

- `INTBASE-Remediation.ps1` — guarded (`-WhatIf` default) fix for a missing auto-enrollment/CA scope

or the missing group owner.

Configuration templates (Plus / We-Deploy variant only)

- `autopilot-deployment-profile.json` — sample user-driven, Entra-joined Autopilot profile.
- `enrollment-status-page.json` — sample minimal Enrollment Status Page config.
- `windows-enrollment-restrictions.json` — sample Windows enrollment platform restriction.
- `autopilot-hardware-hash.csv` — hardware-hash CSV column template for import.

The two ways to buy

	Essentials (Self-Deploy)	Plus (We-Deploy)
Documentation (PDF + Word)	Yes	Yes
PowerShell scripts	—	Yes (5)
Configuration templates	—	Yes (4)
Done-for-you deployment	—	**Yes** — we schedule and run your basic Intune stand-up + first Windows enrollments after checkout
Bundle	`INTBASE-Essentials.zip`	`INTBASE-Complete-Solution.zip`

Objects this solution creates (for your records)

`INTBASE - Windows Users - Pilot` · `INTBASE - Autopilot Devices` (Assigned; owned by the Intune Provisioning Client) · `INTBASE - Autopilot - Pilot` · `INTBASE - ESP - Pilot` · `INTBASE - Windows Enrollment - Pilot` · `INTBASE - Compliance - Windows Pilot`

Where to go next

Free **Enable Microsoft Intune Quick-Start** → **this solution** → **Windows Autopilot** and **Cloud-First Autopilot Zero-Touch** for production-scale depth.

Appendix A — PowerShell Scripts

Included in the accompanying ZIP. Each script's header documents full usage.

INTBASE-Get-AutopilotHash.ps1

Purpose: Captures the local device's Autopilot hardware identity (4K HH) and writes a CSV ready to import into Intune (Devices > Enrollment > Windows > Devices > Import), so the device can be registered for basic Entra-joined Autopilot.

How to run: In an elevated PowerShell window on the device: powershell -ExecutionPolicy Bypass -File .\INTBASE-Get-AutopilotHash.ps1 -OutFile C:\Temp\INTBASE-AutopilotHWID.csv Collect multiple devices by appending to the same CSV with -Append.

INTBASE-Preflight.ps1

Purpose: Read-only preflight for the Microsoft Intune First-Time Setup for Windows. Confirms Intune-bearing licensing exists, the MDM authority is Intune, Windows automatic-enrollment is scoped, and the Intune Provisioning Client service principal owns the Autopilot device group used for enrollment-time grouping.

How to run: Connect-MgGraph -Environment Global -Scopes "Organization.Read.All","DeviceManagemeentServiceConfig.Read.All","Group.Read.All","Directory.Read.All" Government: Connect-MgGraph -Environment USGov (or USGovDoD) .\INTBASE-Preflight.ps1 -AutopilotGroupName "INTBASE - Autopilot Devices"

INTBASE-Remediation.ps1

Purpose: Optionally repair the two settings that most often block first-time Windows enrollment: (1) the Intune Provisioning Client is not an OWNER of the Autopilot device group used for enrollment-time grouping, and (2) Windows automatic-enrollment MDM user scope does not include the pilot group. Defaults to a safe preview (-WhatIf behaviour): it reports what it WOULD change and makes no changes unless -Apply is passed.

How to run: Connect-MgGraph -Environment Global -Scopes "Group.ReadWrite.All","Directory.ReadWrite.All","DeviceManagementServiceConfig.ReadWrite.All" Government: Connect-MgGraph -Environment USGov (or USGovDoD) .\INTBASE-Remediation.ps1 # preview only (default, no changes) .\INTBASE-Remediation.ps1 -Apply # actually add the group owner

INTBASE-Report-Enrollment.ps1

Purpose: Read-only inventory of enrolled Windows devices with their enrollment method (user-driven Entra join vs Windows Autopilot), ownership, management state, and compliance — the evidence that both first-time enrollment paths landed devices under management.

How to run: Connect-MgGraph -Environment Global -Scopes "DeviceManagementManagedDevices.Read.All" Government: Connect-MgGraph -Environment USGov (or USGovDoD) .\INTBASE-Report-Enrollment.ps1 -CsvPath .\INTBASE-Enrollment-Report.csv

INTBASE-Validate-Enrollment.ps1

Purpose: Read-only validation for a single Windows device. Confirms the device is present in Intune, managed (MDM), Microsoft Entra joined, and reporting compliant — the exit criterion for both the user-driven and Autopilot enrollment paths.

How to run: Connect-MgGraph -Environment Global -Scopes "DeviceManagementManagedDevices.Read.All","Device.Read.All" Government: Connect-MgGraph -Environment USGov (or USGovDoD) .\INTBASE-Validate-Enrollment.ps1 -DeviceName "PILOT-PC-01"